Practical No. 3

Aim :- To Practice advaced SQL queries using filters (WHERE), Sorting (ORDER BY), aggregation (GROUP BY), & nested subqueries in a student Course management System.

Theory :-

(1) Filters (WHERE Clause)

- Used to retrieve specific rows from a table based on conditions.
- Operators : =, <, >, <=, >=, != , LIKE, IN, BETWEEN.
- e.g; find students above age 20.

(2) Sorting (ORDER BY Clause)

- Arrange query results in ascending (ASC) or descending (DESC) order.
- e.g; list courses alphabetically or by number of Credits.

(3) Aggregation (GROUP BY Clause)

- Groups rows with the same column values & applies aggregate functions like: COUNT(), AVG(), SUM(), MAX(), MIN().
- e.g; Count number of students enrolled per course.

(4) Nested Subqueries

- A query inside another query.
- Used for complex conditions where results of one query determine another query.
- e.g; find students enrolled in courses with more

than 3 Credits.

Conclusion

- Filters with ~~WE~~ WHERE allow targeted data retrieval.
- Sorting Using ORDER BY makes data analysis easier
- Aggregation with GROUP BY provides summarized information.
- Nested subqueries allow Complex Conditional queries.
- Lab demonstrates advanced SQL querying techniques for efficient reporting in a student Course management System.